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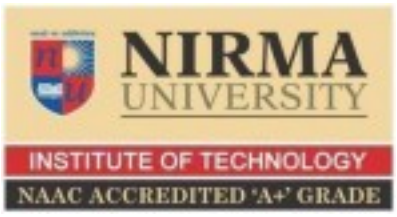
Software Project Management - 4CS504ME25 B.

Tech. Semester IV

Prepared By :

Gandhi Urva Yogeshkumar (23BCE078)

Gajnotar Rakshit (23BCE077)



Institute of Technology,
Nirma University
Ahmedabad - 382481

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Aim

To identify, assess, prioritize and plan response strategies for project, technical and operational risks in the Hospital Management System (HMS), compiling a comprehensive Risk Register.

Objective

- Identify project, technical and security risks from the constraints defined in T1 and T3.
- Perform qualitative risk assessment by rating Probability (1-5) and Impact (1-5) for each risk.
- Map risks onto a 5x5 Probability and Impact Matrix to classify severity as Critical, High, Medium or Low.
- Create preventive mitigation actions and fallback contingency plans for each risk.
- Compile a complete Risk Register with assigned owner roles and priority rankings.
- Establish a weekly risk monitoring procedure to track warning triggers.

1. Introduction

A risk is an uncertain future event that, if it occurs, hurts a project in terms of schedule, cost, quality or reputation. Risk management means identifying these events in advance so the team can prevent them or handle them quickly instead of being caught unprepared.

In T1 we listed the constraints of the Hospital Management System: a small two-student team, a strict 16-week semester timeline, patient medical data privacy, zero software budget and hospital operational reliability. Each constraint creates potential risks. In this tutorial we identify the risks of the HMS project, assess them qualitatively and prepare a complete risk register with response strategies.

The risk management process follows four steps:

- Identify what can go wrong.
- Assess how likely it is and how much damage it would cause (qualitative assessment).
- Plan responses: decide whether to avoid, mitigate, transfer or accept each risk and prepare contingency plans.
- Monitor risks during sprint execution and update the register weekly.

2. Risk Identification

We identified 12 realistic risks for our Hospital Management System across project, technical, security and operational categories:

- R1 (Project): A team member becomes unavailable (exams, placements, illness), leaving heavy workload for one member.
- R2 (Security/Legal): Patient medical data breach or unencrypted health record exposure violating privacy rules.
- R3 (Security): Role-based access control setup flaw allows unauthorized doctor or receptionist data tampering.
- R4 (Project): Scope creep from extra module requests (e.g. ICU bed tracker, OT scheduler) delaying the core timeline.
- R5 (Technical): Database crash or patient data corruption in PostgreSQL/MongoDB without recent backups.
- R6 (Technical): API integration mismatch between Express REST endpoints and React forms.

- R7 (Technical): Race conditions in concurrent appointment booking resulting in double-booked doctor slots.
- R8 (Technical): Pharmacy medicine stock inventory mismatch during simultaneous prescription checkouts.
- R9 (Operational): Free-tier cloud hosting goes to sleep or experiences latency during mentor demos.
- R10 (Operational): SMS notification gateway service limits or delivery failures for appointment tokens.
- R11 (Business): Billing calculation errors when combining appointment fees, doctor charges and pharmacy discounts.
- R12 (Technical): UI layout or table rendering bugs on older clinic computers or low-resolution tablets.

3. Qualitative Risk Assessment

Qualitative risk assessment evaluates each risk by its likelihood (Probability) and severity (Impact):

- Probability Scale: 1 (Very Low, <10%), 2 (Low, 10-30%), 3 (Medium, 30-50%), 4 (High, 50-70%), 5 (Very High, >70%).
- Impact Scale: 1 (Very Low), 2 (Low), 3 (Medium), 4 (High), 5 (Very High: project failure or data loss).
- Risk Exposure Score: Calculated as Probability * Impact, ranging from 1 to 25. Severity levels are: Critical (15-25), High (10-14), Medium (5-9) and Low (1-4).

The assessment scores and rankings for all 12 risks are presented below:

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score (P * I)	Level	Rank
R1	Team member unavailable (exams, illness, placements)	4 (High)	5 (Very High)	20	Critical	1
R3	Role-based access control flaw allowing unauthorized data tampering	3 (Medium)	4 (High)	12	High	2
R4	Scope creep with extra module requests delaying core modules	4 (High)	3 (Medium)	12	High	3
R7	Race condition in appointment booking causing double-booking	3 (Medium)	4 (High)	12	High	4
R2	Patient medical data breach or unencrypted record exposure	2 (Low)	5 (Very High)	10	High	5
R5	Database crash or data corruption with no recent backup	2 (Low)	5 (Very High)	10	High	6
R6	API integration mismatch between Express and React components	3 (Medium)	3 (Medium)	9	Medium	7
R8	Pharmacy inventory mismatch during concurrent checkouts	3 (Medium)	3 (Medium)	9	Medium	8
R11	Billing calculation errors combining appointment and drug discounts	3 (Medium)	3 (Medium)	9	Medium	9
R9	Free cloud hosting sleep or slow response during demo	3 (Medium)	2 (Low)	6	Medium	10
R10	SMS gateway rate limit or delivery failure for appointment tokens	3 (Medium)	2 (Low)	6	Medium	11
R12	UI rendering and table layout issues on older clinic	2 (Low)	2 (Low)	4	Low	12

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score (P * I)	Level	Rank
	screens					

Probability and Impact Matrix

The matrix below maps each risk into its probability and impact coordinates:

Probability \ Impact	1 Very Low	2 Low	3 Medium	4 High	5 Very High
5 Very High	-	-	-	-	-
4 High	-	-	R4	-	R1
3 Medium	-	R9, R10	R6, R8, R11	R3, R7	-
2 Low	-	R12	-	-	R2, R5
1 Very Low	-	-	-	-	-

- Critical Risk: R1 (Team member unavailability) scores 20 because a two-person team has no spare staff. If one student falls sick during exam periods, project progress stops unless work is shared.
- High Risks: R3 (Role access tampering), R4 (Scope creep), R7 (Double booking) and R2/R5 (Data breach and database loss) carry severe consequences and require proactive technical safeguards.
- Medium and Low Risks: Risks like R9 (hosting sleep) and R12 (UI issues) are managed through simple preventive checks and local demo fallbacks.

4. Risk Register

The Risk Register is the core tracking document for project risks. It records each risk, its severity level, response strategy, preventive mitigation actions, contingency fallback plans and the assigned team owner:

ID	Risk Description	Score	Strategy	Mitigation (Preventive Action)	Contingency Plan (If Triggered)	Owner	Rank
R1	Team member unavailable (exams, illness, placement)	20 (Crit)	Mitigate	Share knowledge across all modules; keep modular Git commits; document API routes; keep 10% schedule buffer.	Drop optional dashboard charts; reassign non-critical tasks to active member; reschedule sprint reviews.	Urva	1
R3	Role-based access flaw allows unauthorized data edits	12 (High)	Mitigate	Central role verification middleware on every protected route; write automated route permission tests.	Instantly disable vulnerable route; patch role middleware; review database audit logs.	Urva	2
R4	Scope creep with extra module requests delaying core work	12 (High)	Mitigate	Freeze requirements based on T1-T3; maintain strict backlog; require faculty approval for changes.	Defer extra requests to future roadmap; keep focus strictly on 6 core modules.	Rakshit	3
R7	Race condition in appointment booking causing double-booking	12 (High)	Mitigate	Use database transactions with row-level locks on doctor time slots; validate slot status before commit.	Show double-booking error toast; automatically refresh slot grid with next	Urva	4

ID	Risk Description	Score	Strategy	Mitigation (Preventive Action)	Contingency Plan (If Triggered)	Owner	Rank
					available slot.		
R2	Patient medical data breach or unencrypted record exposure	10 (High)	Avoid	Store passwords with bcrypt; encrypt sensitive medical fields; enforce HTTPS and strict JWT expiration.	Revoke active JWT tokens; force credential reset; notify project mentor immediately.	Urva	5
R5	Database crash or data corruption with no recent backup	10 (High)	Transfer	Automated daily pg_dump database exports; save SQL backup copies in separate private storage.	Restore database from latest daily SQL dump; re-run database migration scripts.	Urva	6
R6	API integration mismatch between Express and React	9 (Med)	Mitigate	Document API request/response formats in Swagger early; run weekly joint integration checks.	Use schedule buffer; fix parameter mismatches together; use mock JSON temporarily.	Rakshit	7
R8	Pharmacy inventory mismatch during concurrent checkouts	9 (Med)	Mitigate	Implement atomic database decrements for medicine quantities with stock validation checks.	Roll back checkout transaction if stock hits zero; alert staff of inventory shortage.	Urva	8
R11	Billing calculation errors combining fees and discounts	9 (Med)	Mitigate	Centralize billing calculation in one backend helper function with automated unit test assertions.	Review invoice formula; manually recalculate disputed bill; patch backend calculation helper.	Rakshit	9
R9	Free cloud hosting sleep or slow response during demo	6 (Med)	Mitigate	Ping cloud endpoints before evaluation; keep fully working local deployment copy on laptops.	Run demo locally using localhost:5000 and local database; show recorded video backup.	Rakshit	10
R10	SMS gateway rate limit or delivery failure for tokens	6 (Med)	Transfer	Track free SMS quota; display appointment token directly on UI confirmation screen.	Rely on on-screen booking confirmation slip and email fallback when SMS fails.	Rakshit	11
R12	UI rendering and table layout issues on older clinic screens	4 (Low)	Accept	Test UI on Chrome, Firefox, Edge and tablet viewports; use responsive CSS flexbox/grid.	Apply quick CSS patch for table scrollbars on low-resolution clinic displays.	Rakshit	12

Response Strategies Used

- Avoid: Change system design to eliminate the risk (R2: encrypt sensitive fields and enforce JWT auth).
- Mitigate: Take concrete steps to lower probability or impact (R1, R3, R4, R6, R7, R8, R9, R11).
- Transfer: Shift operational burden to external tools or backups (R5: automated SQL dumps; R10: cloud SMS gateways).
- Accept: Live with minor risks where the cost of prevention exceeds the impact (R12: responsive layout adjustments).

5. Risk Monitoring Protocol

- Weekly Review: Team reviews the risk register at every weekly sprint meeting, checking whether any risk score has changed.
- Trigger Tracking: Early warning signs are monitored actively (e.g. missed task commits indicating R1, rising CORS errors indicating R6, or inventory discrepancies indicating R8).
- High Risk Priority: Critical and High risks (R1, R3, R4, R7, R2, R5) are reviewed before merging code into main branches.
- Reserve Allocation: The 10% effort reserve established in our T7 COCOMO estimate is reserved specifically for executing contingency actions if risks materialize.

6. Conclusion

Risk management does not eliminate every problem, but it allows the Hospital Management System team to identify major threats early and prepare structured responses. In our project, team availability, medical data privacy, access control and appointment race conditions are the top priorities. Assigning specific owner roles to Urva and Rakshit and reviewing the register weekly keeps risks under control throughout the semester.